

Gamma-ray Analysis for Multi-Messenger Astrophysics: GAMMA

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I. INTRODUCTION

The Gamma-ray Analysis for Multi-Messenger Astrophysics (GAMMA) initiative is a project by the CAPIBARA Collaboration with the goal to study high-energy, gamma-ray phenomena in the universe and launch a CubeSat to observe these events[cite: 2]. In this white paper, we provide a set of compelling open problems that motivate the development and launch of such a mission[cite: 2].

In the 2020 decadal survey [1], it was pointed out that multi-band (across the electromagnetic spectrum) and multi-messenger (combining different cosmic messengers such as light, gravitational waves, and neutrinos) detections were to be prioritized in the coming years[cite: 3]. Currently operating flagship gamma-ray observatories have long passed their design operational lifetimes, and new replacements are still in very early phases[cite: 4]. Professionally and student-developed CubeSats with miniaturized detector technology have proven capable of successfully detecting gamma-ray bursts (GRBs) [2–4][cite: 5].

However, high-energy gamma-ray photons cannot be focused using traditional optical telescope designs[cite: 6]. A highly effective method to accurately observe and locate GRBs is to perform triangulation using a constellation of multiple satellites in orbit[cite: 7]. Therefore, having cooperative missions with wide fields of view and robust resolution capabilities is of paramount importance for the detailed study of these extreme astrophysical phenomena[cite: 8]. Because Earth’s atmosphere completely blocks gamma-ray radiation, space-based detectors are fundamentally required to capture these brief, intense flashes of high-energy light, which originate from catastrophic cosmic events like the core-collapse of massive stars or the coalescence of binary neutron stars[cite: 9].

II. MULTI-MESSENGER ASTRONOMY

III. COSMOLOGY: GRBS AS HIGH- z PROBES

A. Cosmological Models and Tensions

Since the late 1920s, it has been established that the universe is expanding [5, 6], a discovery that led to the formulation of the Big Bang theory, later confirmed by the observation of the Cosmic Microwave Background (CMB) [7, 8][cite: 10]. In the late 1990s, the expansion of the universe was found to be accelerating via Type Ia Supernovae observations [9, 10], motivating the description of the expansion rate as a time-dependent parameter $H(t)$ (the Hubble parameter), with $H_0 \equiv H(t_0)$ denoting its present value[cite: 11]. The relation between the luminosity distance d_L and the cosmological redshift z is given by[cite: 12]:

$$d_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')} \quad (1)$$

where c is the speed of light, H_0 is the Hubble-Lemaître constant, and $E(z)$ is the dimensionless Hubble parameter, which varies depending on the specific cosmological model under consideration[cite: 12].

The current standard model of cosmology, Λ CDM, assumes the dominance of cold dark matter (CDM) and a cosmological constant Λ [?][cite: 13]. Derived from the Friedmann equations, which treat the components of the universe as perfect fluids, the dimensionless expansion

rate is expressed as[cite: 14]:

$$E^2(z) = \left(\frac{H(z)}{H_0} \right)^2 = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda \quad (2)$$

where Ω_i represents the present-day energy density parameters for radiation (Ω_r), matter (Ω_m), spatial curvature (Ω_k), and dark energy (Ω_Λ) [cite: 14, 15]. Typically, a spatially flat universe is assumed ($\Omega_k = 0$) with a negligible radiation contribution ($\Omega_r \approx 0$), leading to the constraint $\Omega_\Lambda + \Omega_m = 1$ [cite: 15]. In baseline reference configurations where cosmological parameters are fixed, standard values such as $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, and $h = 0.7$ (where $H_0 = 100h$ km/s/Mpc) are commonly adopted, in excellent agreement with large-scale structure measurements [cite: 16].

In recent years, considerable attention has focused on the “Hubble Tension,” an unresolved discrepancy where local direct measurements of H_0 using the distance ladder disagree with early-universe CMB observations [11–13] [cite: 17]. In the era of precision cosmology, this conflict—alongside secondary anomalies in the matter fluctuation amplitude σ_8 and the matter density Ω_m [?]—has become increasingly significant [cite: 18]. These tensions, combined with the theoretical fine-tuning problem of Λ [?] and growing tentative evidence for a dynamical dark energy (DDE) component [?], motivate the development of alternative, independent cosmic distance indicators to validate or extend the standard model [cite: 19].

B. GRBs as Cosmological Probes

Gamma-ray bursts are among the most luminous astronomical events in the Universe and are visible at high redshifts, making their potential use as cosmological probes highly compelling [cite: 20]. However, GRBs are not standard candles, and we must rely on empirical relations between observables to compute our luminosity distance and thus test different cosmologies [cite: 21]. Multiple potential relations to “standardize” GRBs have been proposed [?] [cite: 22].

When utilizing GRBs to probe cosmological parameters, we encounter the circularity problem: a specific cosmology $\bar{\Omega}$ must be assumed a priori to compute the intrinsic properties from observed quantities and fit the empirical relation, which is then used to constrain the very same parameters [cite: 23]. For example, the Amati relation connects the intrinsic spectral peak energy $E_{p,i}$ and the isotropic equivalent energy E_{iso} via [cite: 24]:

$$E_{p,i} = a \cdot E_{\text{iso}}^b \quad (3)$$

Here, E_{iso} is computed from the observed bolometric fluence $\mathcal{F}$ relying on the luminosity distance by [cite: 24]:

$$E_{\text{iso}}(\bar{\Omega}) = \frac{4\pi \mathcal{F} d_L^2(z, \bar{\Omega})}{1+z} \quad (4)$$

In this case, in order to compute the calibration parameters (a, b), a baseline cosmology must be adopted [cite: 24]. The circularity problem can be avoided by evaluating the empirical relation fit across a grid of possible cosmologies, where the correct model minimizes the intrinsic scatter within the relation [cite: 25].

Key instrument requirements for this purpose include a broad energy range (typically from 10 keV up to several MeV) to ensure the spectral peak energy $E_{p,i}$ is captured, alongside a sufficient energy resolution ($\Delta E/E \approx 10\text{--}20\%$) to accurately fit the spectrum [cite: 31]. Additionally, a wide field of view is critical to maximize the detection rate of these short-lived transient events, drastically improving the statistical power of the catalog [cite: 32].

According to Fisher matrix forecasts, roughly 800 to 1,000 detections with an average measurement uncertainty of 0.10 dex (corresponding to approximately 25% error on the energy observables) are required to constrain the matter density parameter Ω_m to a 10% precision level using the Amati relation [cite: 33]. Reaching a 1% precision scale demands sample sizes in the tens of thousands due to the hard mathematical floor established by the intrinsic scatter of these relations [cite: 34].

While the Amati relation relies solely on prompt gamma-ray emission, other empirical relations can tighten this intrinsic scatter but pose stricter multi-wavelength requirements [cite: 35]. For instance, the Ghirlanda relation connects the peak energy to the collimated true energy E_γ and features a significantly lower intrinsic scatter (≈ 0.10 dex) [cite: 36]. However, utilizing it requires precise X-ray or optical follow-up to observe the afterglow jet break time and calculate the opening angle θ_j [cite: 37]. Furthermore, because GRBs lack low-redshift anchors, they struggle to independently constrain the Hubble constant H_0 , creating a degeneracy with the empirical relation’s zero-point [cite: 38]. Therefore, GRBs are most powerful when utilized as complementary probes alongside gravitational wave luminosity distance data [cite: 39].

C. Multi-Messenger Cosmology

The integration of gravitational wave (GW) standard sirens and gamma-ray bursts provides an exceptionally powerful multi-messenger framework for breaking cosmological parameter degeneracies [cite: 40]. While standard sirens yield absolute, calibration-free luminosity distances that directly constrain H_0 , they suffer from a strong inclination-distance degeneracy and are currently limited to the relatively local universe ($z \lesssim 0.1$ for binary neutron star mergers). Conversely, GRBs can be observed up to the highest redshifts ($z \gtrsim 9$), making them excellent probes of matter density Ω_m and dark energy evolution $w(z)$ [cite: 20]. By combining the local absolute calibration of GW standard sirens with the high-redshift lever arm of GRB empirical relations, the mutual parameter degeneracies are effectively broken, yielding significantly tighter joint constraints [cite: 30, 39].

D. GRB Statistics Estimations

In order to compute cosmological parameters with accuracy, a large sample of GRBs is needed so that the fit of the empirical relation over the cosmological parameter space has the lowest error uncertainty possible[cite: 41]. We compute the number of GRBs necessary to observe with average measurement uncertainty ($0.01\text{dex} < \sigma_{\text{obs}} < 0.2\text{dex}$) to obtain 10% and 1% precision on Ω_m [cite: 42]. For the error uncertainty propagation, we compare three different standard empirical relations: the Amati, Yonetoku, and Ghirlanda relations [?][cite: 43].

- Amati: $E_{\text{p},i} \propto E_{\text{iso}}^a$ [cite: 44]
- Yonetoku: $E_{\text{p},i} \propto L_{\text{iso}}^b$ [cite: 44]
- Ghirlanda: $E_{\text{p},i} \propto E_{\gamma}^c$ [cite: 44]

Where a, b, c are free parameters, L_{iso} is the isotropic luminosity, which is related to the luminosity distance by $d_L(z) = (L_{\text{iso}}/4\pi F)^{1/2}$, and E_{γ} is the collimation corrected energy, $E_{\gamma} = E_{\text{iso}}(1 - \cos \theta_j)$ with θ_j being the jet opening angle[cite: 44]. E_{γ} can only be computed if θ_j is known, i.e., if the jet break time t_{break} is observed in the afterglow, thus highlighting the importance of rapid alerts and follow-up observations in multiple energy ranges[cite: 45].

Since GRBs are jet-driven phenomena, due to relativistic aberration an observer only sees photons within a cone of aperture defined by $\theta_{\Gamma} \propto \Gamma^{-1}$, where Γ is the bulk Lorentz factor of the emitting material[cite: 46]. This emitting material gets slowed down by the medium around the GRB source; thus, as the jet travels, Γ diminishes and θ_{Γ} increases[cite: 47, 48]. During this time the observer perceives increasing luminosity, since the cone diameter gets larger, followed by a peak and an afterglow as the emission decreases[cite: 49]. The turning point happens when $\theta_{\Gamma} = \theta_j$, at which point the totality of the jet's surface is visible and there is a sudden decrease in luminosity[cite: 50]. If we approximate the afterglow emission as a linear function over time, then the slope becomes smaller[cite: 51].

To compute the estimations on the sample size for each relation, we employ the Fisher information matrix[cite: 52]. For any relation $X - Y$, Y corresponds to a different observable in each relation (see Table I)[cite: 53].

$$\log_{10} Y = a + b \log_{10} E_{\text{p},i} \quad (5)$$

where Y is mathematically related to $d_L(z)$ by[cite: 54]:

$$Y = 4\pi d_L^2(z) \cdot X_{\text{obs}} \cdot (1+z)^{\alpha} \quad (6)$$

Given the distance modulus expression[cite: 54]:

$$\mu(z) = 5 \log_{10} \left(\frac{d_L(z)}{\text{Mpc}} \right) + 25 \quad (7)$$

we substitute $d_L(z)$ into the distance modulus equation to find[cite: 54]:

$$\mu(z) = \frac{5}{2} [a + b \log_{10} E_{\text{p},i} - \log_{10} X_{\text{obs}} - \alpha \log_{10}(1+z) - \log_{10}(4\pi)] + 25 \quad (8)$$

TABLE I. Gamma-Ray Burst Relations

	Relation	Y	X_{obs}	α
Amati	E_{iso}	S_{bolo}	(bolometric fluence)	-1
Ghirlanda	E_{γ}	S_{bolo}	$(1 - \cos \theta_j)$	-1
Yonetoku	L_{iso}	P_{bolo}	(peak flux)	0

Applying standard error propagation while taking into account both the intrinsic physical scatter σ_{int} and the experimental measurement uncertainties yields[cite: 55, 56]:

$$\sigma_{\mu} = \frac{5}{2} \sqrt{\sigma_{\text{int}}^2 + b^2 \sigma_{\log E_{\text{p},i}}^2 + \sigma_{\log X_{\text{obs}}}^2} \quad (9)$$

The intrinsic error represents a fixed, physical variation in the empirical relationship itself[cite: 56]. It ranges from $\sigma_{\text{int}} \approx 0.10\text{dex}$ for the Ghirlanda relation to $\sigma_{\text{int}} \approx 0.30\text{dex}$ for the Amati and Yonetoku relations[cite: 57]. In order to project these errors into the computation of cosmological parameters like Ω_m , we calculate the Fisher Information Matrix F [cite: 58].

For a single distance modulus datum $\mu(z)$, the information contribution regarding Ω_m is given by[cite: 59]:

$$F = \frac{1}{\sigma_{\mu}^2} \left(\frac{\partial \mu}{\partial \Omega_m} \right)^2 \quad (10)$$

Assuming a sample size of N statistically independent GRBs distributed across redshift, the cumulative information matrix scales linearly such that $F_{\text{total}} = N \cdot \langle F \rangle$ [cite: 60]. By setting our target standard error to $\Delta \Omega_m = 1/\sqrt{F_{\text{total}}}$, we can invert the relationship to solve for the necessary sample size N [cite: 61]:

$$N = \frac{\sigma_{\mu}^2}{(\Delta \Omega_m)^2 \left\langle \left(\frac{\partial \mu}{\partial \Omega_m} \right)^2 \right\rangle} \quad (11)$$

The results of our numerical simulations evaluating N against changing average observational uncertainties are presented in Fig. 1[cite: 61]. To achieve a 10% precision constraint on Ω_m , the Ghirlanda relation requires a small sample footprint (~ 27 – 40 GRBs) due to its tight intrinsic scatter[cite: 62]. However, it remains constrained by the subset of events exhibiting visible afterglow jet breaks[cite: 63]. Conversely, the Amati and Yonetoku relations require noticeably larger catalogs (~ 150 – 241 GRBs) to counter their higher 0.30dex intrinsic variance, but offer a more viable path forward due to the ease of obtaining bolometric fluences and peak fluxes from simple light curves[cite: 64].

IV. LORENTZ INVARIANCE AND HIGH-ENERGY PHOTON LAGS

A. A Theory of Quantum Gravity

briefly explain why a theory of quantum gravity is of such relevance



FIG. 1. Estimates of the required sample size of observed GRBs to constrain Ω_m to 10% and 1% precision, computed for the Amati, Yonetoku, and Ghirlanda relations[cite: 65].

B. Detecting Lorentz Invariance with High-Energy Photons

- elaborate on how detecting high-energy photon logs provides proof of a granular (quantized) space-time

- transform the theory into instrumentation requirements (spectral range and resolution, timing resolution)

V. SUMMARY OF TECHNICAL REQUIREMENTS

Section: COSMOLOGY: GRBS AS HIGH- z PROBES

- high enough energy and spectral resolution to accurately identify E_{peak} [cite: 65, 66]
- broad energy range (especially towards higher energies) to don't be blind to E_{peak} [cite: 66]
- broad field of view (to observe more GRBs) [cite: 66]

VI. CONCLUSIONS

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- [1] C. N. Hartman, The Decadal Survey in Astronomy and Astrophysics 2020 (Astro 2020), NSF Award Number 1852611. Directorate for Mathematical and Physical Sciences, Division Of Astronomical Sciences. 2018. (2018).
 - [2] D. Murphy *et al.*, Experimental Astronomy **52**, 10.1007/s10686-021-09779-9.
 - [3] J. Ripa *et al.*, Grbalpha, vzlusat-2 and grbbeta – grb observations with cubesats.
 - [4] F. Fiore *et al.*, in *Space Telescopes and Instrumentation 2020: Ultraviolet to Gamma Ray* (SPIE).
 - [5] E. Hubble, Proceedings of the National Academy of Sciences **15**, 168 (1929), <https://www.pnas.org/doi/pdf/10.1073/pnas.15.3.168>.
 - [6] G. Lemaître, Annales de la Société Scientifique de Bruxelles **47**, 49 (1927).
 - [7] A. A. Penzias and R. W. Wilson, Astrophys. J. **142**, 419 (1965).
 - [8] R. H. Dicke, P. J. E. Peebles, P. G. Roll, and D. T. Wilkinson, Astrophys. J. **142**, 414 (1965).
 - [9] A. G. Riess *et al.*, The Astronomical Journal **116**, 10.1086/300499 ().
 - [10] S. Perlmutter *et al.*, **517**, 565 (1999).
 - [11] E. Di Valentino *et al.*, **38**, 153001 (2021).
 - [12] A. G. Riess *et al.*, The perfect host: Jwst cepheid observations in a background-free sn ia host confirm no bias in hubble-constant measurements ().
 - [13] M. Tristram *et al.*, Astronomy & Astrophysics **682**, 10.1051/0004-6361/202348015.